Final Year Project Proposal
BS Computer Science

IoT Smart Agriculture Monitoring and Controlling System with AI

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1. Introduction

Agriculture is the backbone of Pakistan's economy, but farmers often face problems due to lack of real-time monitoring of soil and environmental conditions. Decisions about irrigation and fertilizer are mostly made through experience rather than data, leading to resource wastage and lower crop yields.

Tobacco is an important cash crop in Khyber Pakhtunkhwa but requires careful management of soil conditions. While many IoT projects exist for crops like rice and wheat, there has been little focus on tobacco-specific solutions.

Our project proposes a **low-cost IoT-based system** that helps farmers monitor their field, receive alerts, and get basic recommendations for irrigation and fertilizer needs. The system will work in **offline (Bluetooth)** and **online (WiFi/Cloud)** modes. Depending on available time and resources, additional AI-based features may be added to make the system smarter and more unique.

2. Aim and Objectives

Aim

To build an IoT-based smart agriculture system for tobacco crop monitoring, with offline alerts and online AI-based recommendations.

Objectives

- Design a low-cost IoT setup using ESP32 and environmental sensors.
- Develop a mobile application (Flutter) for farmers with offline + online functionality.
- Provide basic recommendations based on optimum crop conditions.
- Integrate AI models for improved recommendations if time and resources allow.
- Collect and use local data to make the system relevant for Pakistani farmers.
- Keep the system scalable for other crops in the future.

3. Services Provided by the System

- Real-time monitoring of soil/environment parameters.
- Basic irrigation and fertilizer recommendations by comparing sensor data with optimum values.
- Alerts and notifications to farmers (offline & online).
- Online advanced analysis (if internet available).
- Data storage for trend analysis and learning.

4. Background and Related Work

- Many IoT agriculture systems having one or two sensors and also require continuous internet, which is not always possible in Pakistan.
- Most projects are generic and not crop-specific.
- Our project focuses on **tobacco** and supports **offline + online dual mode**, which makes it different.

5. Learning Outcomes

Students will learn:

- IoT hardware integration (ESP32, sensors).
- Mobile app development with Flutter.
- AI/ML basics applied to agriculture.
- System design with offline + online layers.
- Practical knowledge of data collection and validation.

6. Proposed Methodology

1. Requirement analysis and literature review.
2. Hardware setup and calibration (ESP32 + sensors).
3. Mobile app development (offline sync via Bluetooth, online sync via WiFi/cloud).
4. Basic rule-based recommendations (based on optimum crop conditions).
5. AI/ML model integration (if time permits, for smarter recommendations).
6. System testing and farmer feedback.

7. Scope and Limitations

In Scope

- IoT-based monitoring of soil/environmental conditions.
- Offline + online recommendations.
- Simple rule-based decision support.
- Mobile app for alerts and farmer interface.

Out of Scope(optional)

- Advanced plant disease detection (may be added later if time/resources allow).
- Large-scale automation systems.
- Multi-crop coverage (only tobacco for this FYP).

8. System Architecture (Layered)

- **Hardware Layer:** ESP32 + sensors (moisture, pH, EC, temperature/humidity).
- **Communication Layer:** Bluetooth (offline), WiFi (online).
- **Application Layer:** Flutter app.
- **Cloud Layer:** Database + AI models
- **User Layer:** Farmer

9. Data Sources

- Published agriculture guidelines (tobacco-specific).
- Pakistan Tobacco Board reports.
- Real time Tabacco data (collected during testing).
- International datasets (e.g., PlantVillage)

10. Expected Outcomes

- A working prototype of a smart agriculture system for tobacco crop.
- Basic recommendations for irrigation and fertilizer.
- Offline + online functionality demonstrated.
- Mobile app for farmer interaction.
- AI features for recommendation and alerts

11. Timeline (10–12 months)

Phase	Duration	Tasks
Phase 1	1 month	Requirement analysis & dataset collection
Phase 2	1.5 months	Hardware setup & calibration
Phase 3	2 months	Mobile app development
Phase 4	2.5 months	AI based recommendation and alerts
Phase 5	2 months	Cloud integration
Phase 6	2 months	System testing with real time data
Phase 7	1 month	Documentation & report

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12. References

- FAO Climate-Smart Agriculture Reports.
- IEEE/Elsevier research papers on IoT in agriculture.
- Pakistan Tobacco Board publications.
- PlantVillage dataset.
- Consultations with agricultural experts.